

THEORY OF AUTOMATA ASSIGNMENT #18-

Section 1 :-

Q1).

Ans:- (a). Regular Expression :- $(a+b)(a+b)(a+b)^*$

Language(L) = {aaa, aba, bbb, bab, aab, baa, bba, abb, ...}

So aaa + aba + bbb + bab + aab + baa + bba + abb

aaa + aba + aab + abb + bbb + bab + baa + bba

aa(a+b) + ab(a+b) + ba(b+a) + ba(a+b)

$(a+b)(aa+ab+bb+ba)$

$(a+b)(a(a+b)+b(b+a)) = \underbrace{(a+b)(a+b)(a+b)}_{\text{length 3}} (a+b)^*$ ^{Ans: 3} _{so put (a+b)* to give length ≥ 3}

(b). Regular Expression :- $(a+b)^* a (a+b)^* a (a+b)^*$

Ans:-

(c). Regular Expression :- $b^* + (b^* a b^*) + (b^* a b^* a b^*)$

Ans:-

(d). Regular Expression :- $(a+b)^* bb (a+b)^* bb (a+b)^*$

Ans:-

(e). Regular Expression :- $a^* (a^* b a^* b a^*)^*$

Ans:-

(f). Regular Expression :- $a^* (b^* (aa)^* a^*)^* b^* a^*$

Ans:-

(g). Regular Expression :- $a(a+b)^* b + b(a+b)^* a$

Ans:-

(h). Regular Expression :- $(b + a(ab)^*)^*$

Ans:-

(i). Regular Expression :- $(a+b)^* (aa+bb+aabb)(a+b)^*$

Ans:-

(j). Regular Expression :- $((aa+bb)^* (ab+ba) (aa+bb)^*)^*$

Ans:-

Q2. (a). REG-EX :- $^{\wedge}[0-9]\{a\}[A-ZA-Z]^+ \backslash [A-ZA-Z]@myEmailService.com \$$

Ans:-

(b). REG-EX :- $^{\wedge}(?=.\{1,2\})(?=.\{A-Za-z0-9\})[A-Z].\{1,3\} \$$

Ans:-

(c). REG-EX :- $^{\wedge}(?=.\{b(\text{comp})\})(?=.\{z.\{0\}.\{virus\}\}) \$$

Ans:-

Section 2 :-

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Q1).

(a). Initial state :- $m_1 = q_1$, $m_2 = q_1$.

(b). Final states :- $m_1 = q_2$, $m_2 = q_1, q_4$.

Ans:-

(c). (i). Sequence of states :-

Ans:- (aabb) $m_1 = \rightarrow q_1 \xrightarrow{a} q_2 \xrightarrow{a} q_3 \xrightarrow{b} q_1 \xrightarrow{b} q_1$

For (aabb) $m_2 = \rightarrow q_1 \xrightarrow{a} q_1 \xrightarrow{a} q_1 \xrightarrow{b} q_2 \xrightarrow{b} q_4$

(ii). Sequence of states :-

For (bbaa) $m_1 = \rightarrow q_1 \xrightarrow{b} q_1 \xrightarrow{b} q_1 \xrightarrow{a} q_2 \xrightarrow{a} q_3$

For (bbaa) $m_2 = \rightarrow q_1 \xrightarrow{b} q_2 \xrightarrow{a} q_4 \xrightarrow{a} q_3 \xrightarrow{a} q_2$

(d).

Ans:- Machine acceptance :-

For (aabb) $m_1 = \text{No}$

For (aabb) $m_2 = \text{Yes}$.

(e).

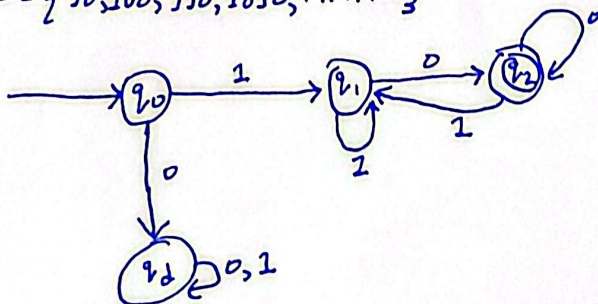
Ans:- For (E) $m_1 = \text{No}$

For (E) $m_2 = \text{Yes}$

Q2). (a).

Ans:-

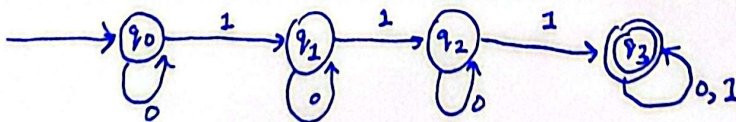
$L = \{ 10, 100, 110, 1010, \dots \}$



(b).

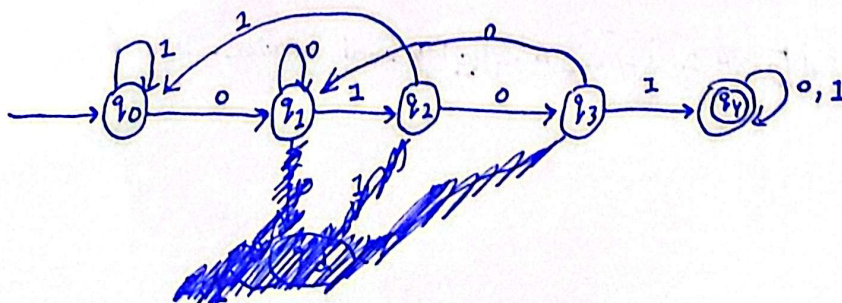
Ans:-

$L = \{ 111, 1111, 1110, 0101010, \dots \}$



(c).

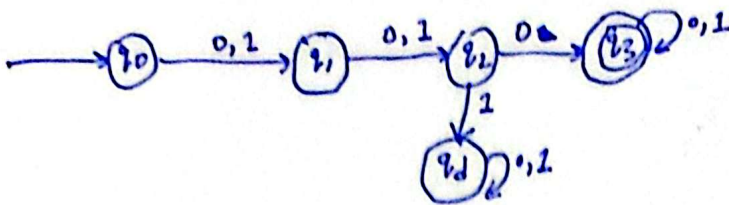
Ans:-



Section 2:-

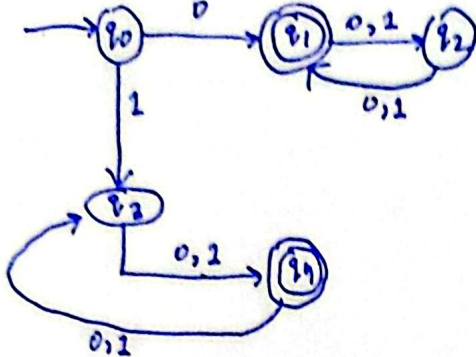
Q2)(d).

Ans:-



Q2)(e).

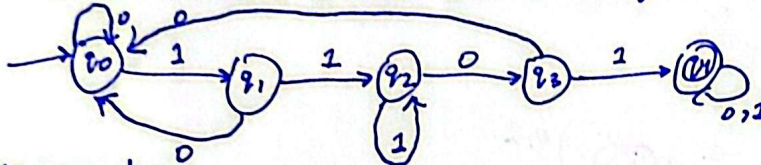
Ans:-



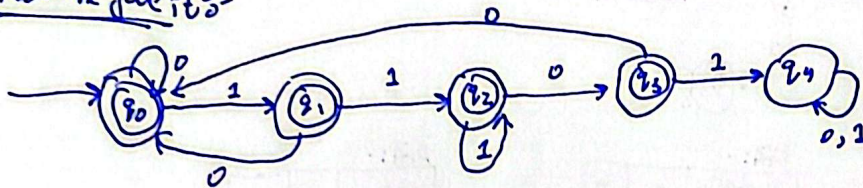
Q2)(f).

Ans:-

As 0's states doesn't contain substring 1101
but we will make first for 1101 containing it.

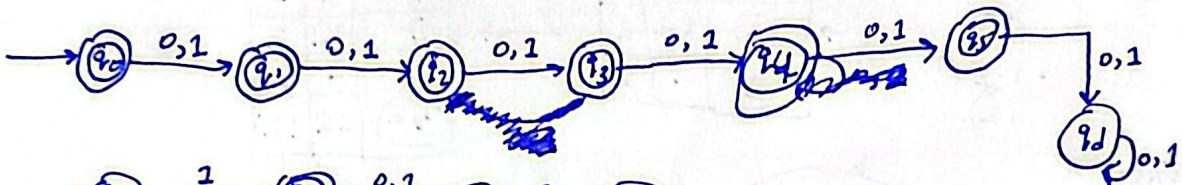


Now negate it:-



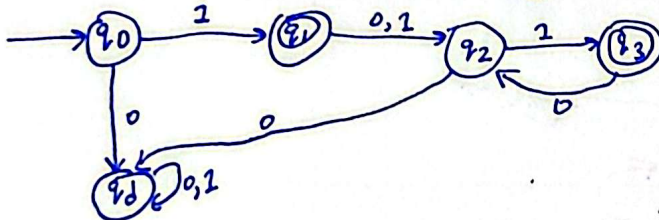
Q2)(g).

Ans:-



Q2)(h).

Ans:-

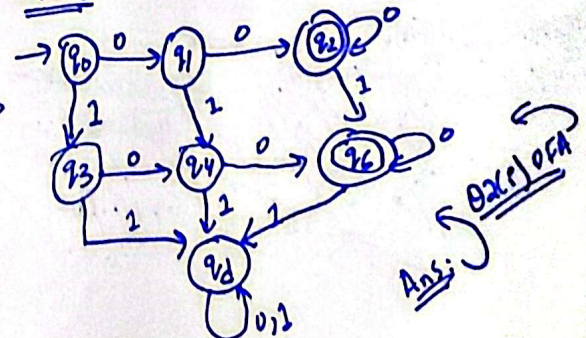
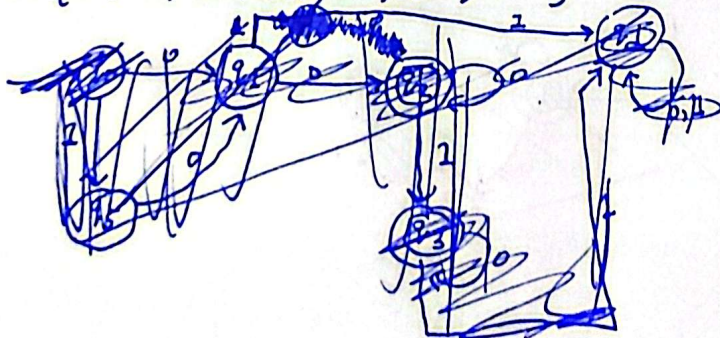


Q2)(i).

Ans:-

$L = \{00, 001, 100, 0001, 1000, 0100, \dots\}$

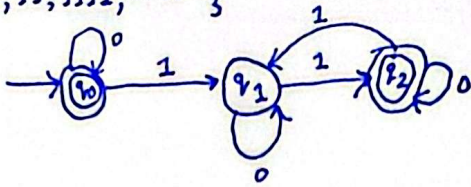
DFA:-



Ans: Q2)(i) DFA

Q2). (j).

Ans:- $\{ \epsilon, 11, 1111, \dots \}$



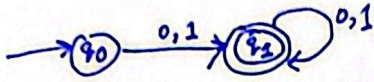
Q2). (k).

Ans:-



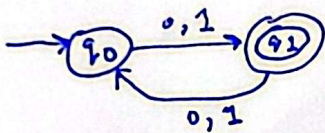
Q2). (l).

Ans:-



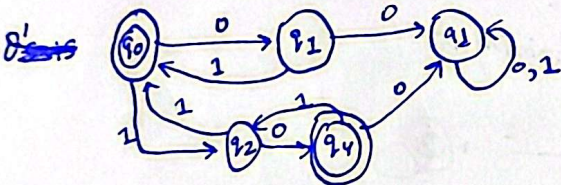
Q2). (m).

Ans:-



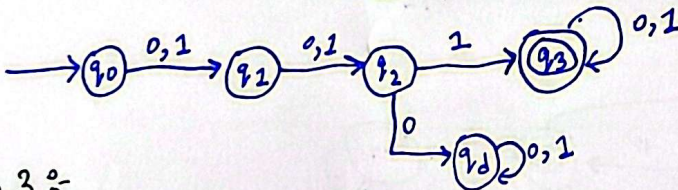
Q2). (n).

Ans:-



Q2). (o).

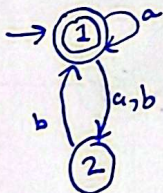
Ans:-



Section 3 :-

Q1). (a). NFA:-

Ans:-



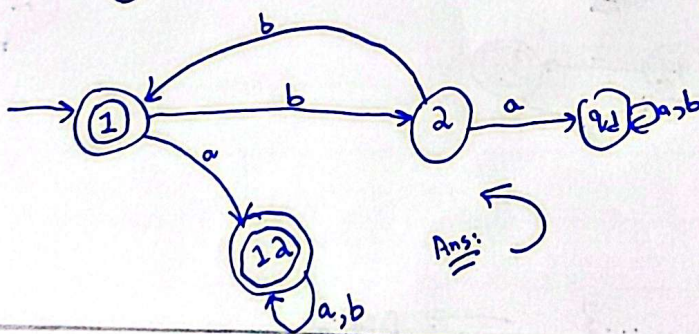
NFA:-

states	a	b
1	1 2	2
2	-	1

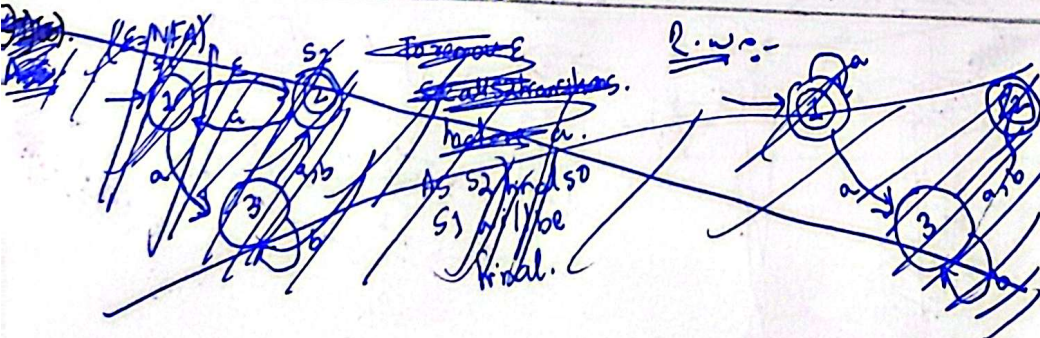
DFA:-

states	a	b
1	1 2	2
2	-	1
1, 2	1 2	1 2

PFA:-



Ans:-



Section 3:-

Q1).

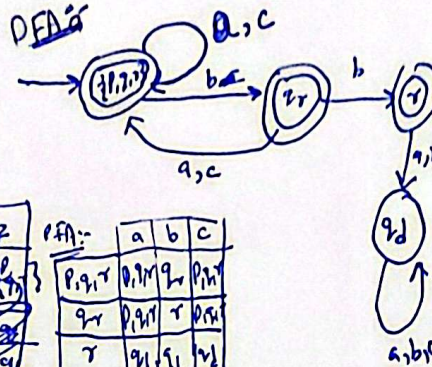
(c). E-NFA:-

Ans:-

For E-NFA:-

	a	b	c	ϵ
$\rightarrow P$	$\emptyset$	q	r	q, r
q	P	r	q	$\emptyset$
$\uparrow r$	$\emptyset$	$\emptyset$	$\emptyset$	$\emptyset$

DFA:-



For DFA:-

	a	b	c
P, q, r	P, q, r	q	P, r
q, r	P, q, r	r	P, r
r	q, r	q	r
q, r	q, r	q, r	r

Q1). (b).

Ans:-

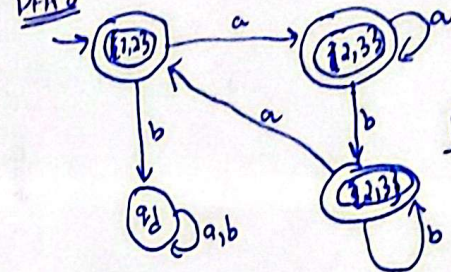
For ENFA:-

	a	b	ϵ
$\rightarrow 1$	2	$\emptyset$	2
2	1	$\emptyset$	$\emptyset$
3	2	$2, 3$	$\emptyset$

DFA:-

	a	b
$1, 2, 3$	$1, 2, 3$	q_d
$1, 2, 3$	$1, 2, 3$	$1, 2, 3$
$2, 3$	$2, 3$	$2, 3$
q_d	q_d	q_d

DFA:-



Ans:-

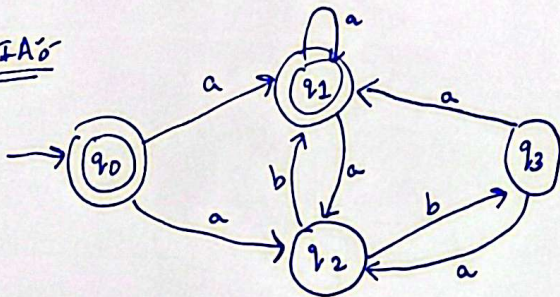
ϵ -closure of (1) = $1, 2$
 " " of (2) = 2
 " " of (3) = 3 .

Q1).

(d).

Ans:-

NFA:-



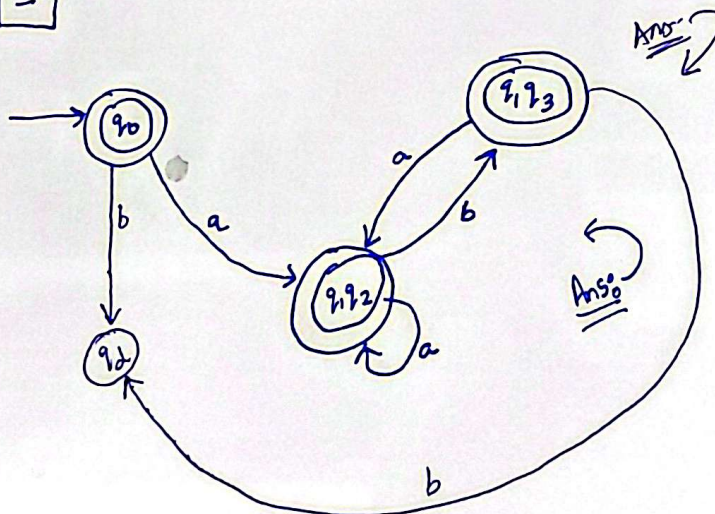
NFA:

state	a	b
q_0	q_1, q_2	-
q_1	q_1, q_2	-
q_2	-	q_1, q_3
q_3	q_1, q_2	-

DFA:-

State	a	b
q_0	q_1, q_2	-
q_1, q_2	q_1, q_2	q_1, q_3
q_1, q_3	q_1, q_2	-

DFA:-



Section 4^o-

Ans:-

